Shabnam Hakimi, Ph.D.

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Education

Caltech — Pasadena, CA
Ph.D., Computation and Neural Systems — 2014

Stanford University — Stanford, CA
B.A., Psychology — 2006

Professional Experience

Toyota Research Institute — Los Altos, CA
Senior Research Scientist, Human-Centered AI — 01/2021 – Present

Duke University — Durham, NC
Postdoctoral Scholar, Center for Cognitive Neuroscience — 06/2016 – 01/2021

University of Colorado, Boulder — Boulder, CO
Postdoctoral Research Associate, Institute of Cognitive Science — 04/2015 – 05/2016

Welltok, Inc. — Denver, CO
Director, CaféWell Behavioral Labs — 08/2013 – 01/2015

California Institute of Technology — Pasadena, CA
Graduate Research and Teaching Fellow, Computation and Neural Systems — 09/2008 – 08/2013

National Institute of Mental Health (NIH) — Bethesda, MD
Intramural Research Training Award Fellow, National Institute of Mental Health (NIMH) — 08/2006 – 07/2008

Stanford University — Stanford, CA
Research Assistant, Department of Psychology — 05/2003 – 06/2006

Grants & Awards

National Institute of Mental Health, NIH — Bethesda, MD
Ruth Kirchstein National Research Service Award (NRSA) — 2017 – 2020

- Project: “Modeling the neurobehavioral basis of extrinsic and volitional motivation in ADHD”
- Role: PI (F32 MH114608)

National Institute of Aging, NIH — Bethesda, MD
Scientific Research Network on Decision Neuroscience & Aging (SRNDNA) Pilot Grant — 2019 – 2020

- Project: “Neurobehavioral correlates of volitional regulation of motivation across the lifespan”
- Role: Subaward PI (R24 AG054355 to G.R. Samanez-Larkin)

Duke University — Durham, NC
SciComm Fellow, Duke Initiative for Science and Society — 2016 – 2017

California Institute of Technology — Pasadena, CA
NSF Integrative Graduate Education and Research Traineeship (IGERT) Fellow — 2008 – 2013

National Institute of Mental Health, NIH — Bethesda, MD
Intramural Research Training Award — 2006 – 2008

University of Washington eScience Institute — Seattle, WA
Neurohackweek (Neurohackacademy) Fellow — 2016

NeuroLeadership Institute — New York, NY
Application of Science Award Runner-Up — 2016

University of California, Santa Barbara — Santa Barbara, CA
Summer Institute in Cognitive Neuroscience Fellow — 2009

California Institute of Technology — Pasadena, CA
NSF/Caltech IGERT fMRI Award — 2009

University of Wisconsin Symposium on Emotion — Madison, WI
HealthEmotions Research Institute Symposium Scholar — 2008

University of Wisconsin Symposium on Emotion — Madison, WI
HealthEmotions Research Institute Symposium Fellow — 2007

University of Wisconsin Symposium on Emotion — Madison, WI
HealthEmotions Research Institute Symposium Scholar — 2006

Stanford University — Stanford, CA
Undergraduate Research Programs Conference Travel Grant — 2006

Stanford University — Stanford, CA
Vice Provost for Undergraduate Education Summer Research Grant — 2005

University of Wisconsin Symposium on Emotion — Madison, WI
HealthEmotions Research Institute Symposium Scholar — 2005

Stanford University — Stanford, CA
Vice Provost for Undergraduate Education Summer Research Grant — 2004

Publications

** equal contributions, shared first authorship | † mentee co-author*

Peer-Reviewed Journal Publications

[1] Paredes, P. E., **Hakimi, S.**, Hong, M. K., Mora-Mendoza, M. A., Murakami, K., Bravo, N. S., Cogny, A., and Klenk, M. "Unstuck - Ubiquitous just-in-time interventions to unleash creativity and sustain wellbeing of creative professionals". In: *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 10.2 (June 2026). DOI: 10.1145/3810227.

[2] Sumner, E. S., DeCastro, J., Costa, J., Gopinath, D. E., Kimani, E., **Hakimi, S.**, Morgan, A., Best, A., Nguyen, H., Brooks, D. J., ul Haq, B., Patrikalakis, A., Yasuda, H., Sieck, K., Balachandran, A., Chen, T. L., and Rosman, G. "Personalizing driver safety interfaces via driver cognitive factors inference". In: *Scientific Reports* 14.1 (2024), p. 18058. DOI: 10.1038/s41598-024-65144-8.

[3] Castrellon, J. J., **Hakimi, S.**, Parelman, J. M.†, Yin, L., Law, J. R., Skene, J. A. G., Ball, D. A., Malekpour, A., Beskind, D. H., Vidmar, N., Pearson, J. M., Skene, J. H. P., and Carter, R. M. "Social cognitive processes explain bias in juror decisions". In: *Social Cognitive and Affective Neuroscience* 18.1 (2022), nsac057. DOI: 10.1093/scan/nsac057.

[4] Castrellon, J. J., **Hakimi, S.**, Parelman, J. M.†, Yin, L., Law, J. R., Skene, J. A. G., Ball, D. A., Malekpour, A., Beskind, D. H., Vidmar, N., Pearson, J. M., Carter, R. M., and Skene, J. H. P. "Neural support for contributions of utility and narrative processing of evidence in juror decision making". In: *Journal of Neuroscience* 42.40 (2022), pp. 7624–7633. DOI: 10.1523/JNEUROSCI.2434-21.2022.

[5] **Hakimi, S***, Sinclair, AH*†, Stanley, M, Adcock, RA, & Samanez-Larkin, GR. "Pairing facts with imagined consequences improves pandemic-related risk perception". In: *Proceedings of the National Academy of Sciences* 118.32 (2021), e2100970118. DOI: 10.1073/pnas.2100970118.

[6] Sinclair, A. H.†, Stanley, M. L., **Hakimi, S.**, Cabeza, R., Adcock, R. A., and Samanez-Larkin, G. R. "Imagining a personalized scenario selectively increases perceived risk of viral transmission for older adults". In: *Nature Aging* 1 (2021), pp. 677–683. DOI: 10.1038/s43587-021-00095-7.

[7] **Hakimi, S***, Wong, CA*, Santanam, TS, Madanay, F, Fridman, I, Ford, C, Patel, M, & Ubel, PA. "Applying behavioral economics to improve adolescent and young adult health: A developmentally-sensitive approach". In: *Journal of Adolescent Health* 67.2 (2020), pp. 174–181. DOI: 10.1016/j.jadohealth.2020.10.007.

[8] Botvinik-Nezer, R., Holzmeister, F., Camerer, C. F., Dreber, A., Huber, J., Johannesson, M., Kirchler, M., Iwanir, R., Mumford, J. A., others, including Hakimi, S., Nichols, T. E., Poldrack, R. A., and Schonberg, T. "Variability in the analysis of a single neuroimaging dataset by many teams". In: *Nature* 582 (2020), pp. 84–88. DOI: 10.1038/s41586-020-2314-9.

[9] Mosner, M. G., McLaurin, R. E., Kinard, J. L., **Hakimi, S.**, Parelman, J.†, Shah, J. S., Bizzell, J., Greene, R. K., Cernasov, P. M., Walsh, E., Addicott, M. A., Eisenlohr-Moul, T., Carter, R. M., and Dichter, G. S. "Neural mechanisms of reward prediction error in Autism Spectrum Disorder". In: *Autism Research and Treatment* 2019 (2019), pp. 1–10. DOI: 10.1155/2019/5469191.

[10] Wang, S., Fan, S., Chen, B., **Hakimi, S.**, Paul, L. K., Zhao, Q., and Adolphs, R. "Revealing the world of autism through the lens of a camera". In: *Current Biology* 26.20 (2016), R909–R910. DOI: 10.1016/j.cub.2016.08.055.

[11] **Hakimi, S*** & Hare, TA*. "Enhanced neural responses to imagined primary rewards predict reduced monetary temporal discounting". In: *Journal of Neuroscience* 35.38 (2015), pp. 13103–13109. DOI: 10.1523/JNEUROSCI.1863-15.2015.

[12] **Hakimi, S***, Hare, TA*, & Rangel, A. "Activity in dlPFC and its effective connectivity to vmPFC are associated with temporal discounting". In: *Frontiers in Neuroscience* 8 (2014), p. 50. DOI: 10.3389/fnins.2014.00050.

[13] Zink, C. F., Kempf, L., **Hakimi, S.**, Rainey, C. A., Stein, J. L., and Meyer-Lindenberg, A. "Vasopressin modulates social recognition-related activity in the left temporoparietal junction in humans". In: *Translational Psychiatry* 1.4 (2011), e3–5. DOI: 10.1038/tp.2011.2.

[14] Tost, H., Kolachana, B., **Hakimi, S.**, Lemaitre, H., Verchinski, B. A., Mattay, V. S., Weinberger, D. R., and Meyer-Lindenberg, A. "Common alleles in *OXTR* impact prosocial temperament and human hypothalamic-limbic structure and function". In: *Proceedings of the National Academy of Sciences* 107.31 (2010), pp. 13936–13941. DOI: 10.1073/pnas.1003296107.

[15] Tost, H., Braus, D. F., **Hakimi, S.**, Ruf, M., Vollmert, C., Hohn, F., and Meyer-Lindenberg, A. "Acute D_2 receptor blockade induces rapid, reversible remodeling in human cortical-striatal circuits". In: *Nature Neuroscience* 13.8 (2010), pp. 920–922. DOI: 10.1038/nn.2572.

[16] Zink, C. F., Stein, J. L., Kempf, L., **Hakimi, S.**, and Meyer-Lindenberg, A. "Vasopressin modulates medial prefrontal cortex-amygdala circuitry during emotion processing in humans". In: *Journal of Neuroscience* 30.20 (2010), pp. 7017–7022. DOI: 10.1523/JNEUROSCI.4899-09.2010.

[17] Goldin, P. R., Manber, T., **Hakimi, S.**, Canli, T., and Gross, J. J. "Neural bases of social anxiety disorder: Emotional reactivity and cognitive regulation during social and physical threat". In: *Archives of General Psychiatry* 66.2 (2009), pp. 170–180. DOI: 10.1001/archgenpsychiatry.2008.525.

Preprints & Working Papers

[1] Eum, B., Daviet, R., **Hakimi, S.**, and Webb, R. "When do physiological measures forecast market outcomes? A random utility framework". https://doi.org/10.2139/ssrn.7473478. 2026. DOI: 10.2139/ssrn.7473478.

[2] Mohan, S., Hong, M., Shu, D., Fransen, A., **Hakimi, S.**, and Klenk, M. "Learning heterogeneous preferences". https://doi.org/10.48550/arXiv.2609.17847. 2026. DOI: 10.48550/arXiv.2609.17847.

[3] Kim, T., Hong, M. K., Chen, Y.-Y., Li, J. Q., Van, M. P., **Hakimi, S.**, Kay, M., and Klenk, M. "Personagram: Bridging personas and product design for creative ideation with multimodal LLMs". Under review. https://arxiv.org/abs/2602.06197. 2026. DOI: 10.48550/arXiv.2602.06197.

[4] Hong, M. K., Li, J., Filipowicz, A., Van, M., Murakami, K., Chen, Y.-Y., Mohan, S., **Hakimi, S.**, and Klenk, M. "Deconstructing taste: Toward a human-centered AI framework for modeling consumer aesthetic perceptions". Under review. https://arxiv.org/abs/2601.17134. 2026. DOI: 10.48550/arXiv.2601.17134.

[5] Chen, Y.-Y., Weng, Y., Filipowicz, A., Iliev, R., Chen, F., **Hakimi, S.**, Zhang, Y., Lee, M., Lyons, K., and Wu, C. "Learning to represent individual differences for choice decision making". https://doi.org/10.48550/arXiv.2503.21704. 2025. DOI: 10.48550/arXiv.2503.21704.

[6] Knutson, B., Christiano, D. X.†, Wu, C., Srirangarajan, T., Deng, W., Klenk, M., Wu, C., and **Hakimi, S.** "Brain activity forecasts changing market demand for innovative vehicles". Under review. https://osf.io/xd9mk/. 2024.

[7] Wright, R. N.†, **Hakimi, S.**, LaBar, K., and Adcock, R. A. "The Motivation Beliefs Questionnaire (MBQ): Measuring beliefs about the controllability, usefulness, and goal-specificity of motivational states". Under review. https://osf.io/k8dy4/. 2024.

[8] Sinclair, A. H.†, Hsiung, A.†, Wright, R. N.†, **Hakimi, S.**, and Adcock, R. A. "Pausing to reflect during news consumption counteracts negativity biases in memory". https://doi.org/10.31234/osf.io/jsyeb_v2. 2024. DOI: 10.31234/osf.io/jsyeb_v2.

[9] Filipowicz, A., Carter, S., Bravo, N., Iliev, R., **Hakimi, S.**, Shamma, D. A., Lyons, K., Hogan, C., and Wu, C. "Visual elements and cognitive biases influence interpretations of trends in scatter plots". https://doi.org/10.48550/arXiv.2310.15406. 2023. DOI: 10.48550/arXiv.2310.15406.

[10] DeCastro, J. A., Gopinath, D., Rosman, G., Sumner, E., **Hakimi, S.**, and Stent, S. "Learning latent traits for simulated cooperative driving tasks". https://doi.org/10.48550/arXiv.2207.09619. 2022. DOI: 10.48550/arXiv.2207.09619.
[11] Hsiung, A.†, Pearson, J. M., Poh, J.-H., **Hakimi, S.**, Adcock, R. A., and Huettel, S. A. "Between heuristics and optimality: Flexible integration of cost and evidence during information sampling". https://doi.org/10.1101/2022.05.17.492355. 2022. DOI: 10.1101/2022.05.17.492355.

Peer-Reviewed Conference Proceedings

[1] Chong, L., Hong, M. K., **Hakimi, S.**, Van, M., Klenk, M., and Paredes, P. E. "Novel or surprising? Toward generating wowing designs with stable diffusion". In: *Proceedings of the ASME 2026 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE 2026)*. Aug. 2026.
[2] Nath, S. S.†, Jayawardana, V., Van, M., Klenk, M., and **Hakimi, S.** "Designing rewards for rewarding designs: Demonstrating the impact of rewards on the creative design process". In: *Design Computing and Cognition (DCC'26)*. Paris, France, July 2026. DOI: 10.48550/arXiv.2604.26083.
Best Paper Award (Honorable Mention); Talk.
[3] Zhang, L., Chong, L., Hong, M. K., **Hakimi, S.**, Filipowicz, A. L. S., Klenk, M., and Paredes, P. E. "Wow, Now I See It! — Leveraging the physiology of surprise to help designers uncover desirable generative designs". In: *Creativity and Cognition (C&C '26)*. June 2026, pp. 397–411. DOI: 10.1145/3803784.3807551.
[4] Klenk, M., **Hakimi, S.**, Hong, M., Paredes, P., and Chen, Y.-Y. "Reassessing the value of creative activity traces for understanding and supporting creative work". In: *Herding CATs: Making Sense of Creative Activity Traces Workshop at CHI 2026*. 2026. URL: papers/Klenk2026cats.pdf.
[5] Eum, B., Knutson, B., Daviet, R., **Hakimi, S.**, and Webb, R. "AI-driven interpretable visual features for demand neuroforecasting". In: *2025 Conference on Artificial Intelligence, Machine Learning, and Business Analytics*. Dec. 2025.
[6] **Hakimi, S.**, Van, M., Hong, M. K., Murakami, K., Paredes, P. E., and Klenk, M. "Psychologically-inspired generative AI videos for supporting creativity". In: *2025 International Conference on Computational Creativity (ICCC'25)*. June 2025, pp. 191–200. URL: https://computationalcreativity.net/iccc25/wp-content/uploads/papers/iccc25-hakimi2025psychologically.pdf.
[7] **Hakimi, S***, Nandy, A*†, Van, M, Goucher-Lambert, K, & Klenk, M. "Semantic properties of abstract prompts shape sequential decision making in design". In: *2025 Multidisciplinary Conference on Reinforcement Learning and Decision Making*. June 2025. URL: papers/Hakimi2025semantic.pdf.
[8] Paredes, P. E., Hong, M. K., **Hakimi, S.**, and Klenk, M. E. "Creative commuter: Towards designing moments for idea generation and incubation during the commute". In: *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. Sept. 2024, pp. 51–55. DOI: 10.1145/3641308.3685022.
[9] **Hakimi, S***, Chen, Y-Y*, Van, M*, Chen, F, Hong, M, Klenk, M, & Wu, C. "Understanding the cognitive complexity in language elicited by product images". In: *ICML 2024 Workshop on LLMs and Cognition*. https://arxiv.org/abs/2409.16521. 2024. DOI: 10.48550/arXiv.2409.16521.
[10] Hong, M. K., Paredes, P., **Hakimi, S.**, Van, M., and Klenk, M. "Unstuck: Charting the design space of generative AI-based creativity interventions". In: *Generative AI and HCI (GenAICHI) Workshop at CHI 2024*. 2024. URL: https://generativeaiandhci.github.io/papers/2024/genaichi2024_14.pdf.
[11] Nandy, A.†, Van, M., Li, J., Goucher-Lambert, K., Klenk, M., and **Hakimi, S.** "Semantic properties of word prompts shape design outcomes: Understanding the influence of semantic richness and similarity". In: *International Conference on Design Computing and Cognition*. Cham: Springer Nature Switzerland, July 2024, pp. 241–258. DOI: 10.1007/978-3-031-71922-6_16.
Best Paper Award.
[12] **Hakimi, S.**, Chen, Y., Van, M., Carter, S., Sumner, E., Bravo, N. B., Murakami, K., Zhang, Y., Wu, C. C., and Klenk, M. "Machine learning-based measure of cognitive complexity explains variance in rank-ordered preference". In: *Proceedings of the Annual Meeting of the Cognitive Science Society*. Vol. 45. 2023. URL: https://escholarship.org/uc/item/79g3t154.
[13] Hong, M. K., **Hakimi, S.**, Chen, Y., Toyoda, H., Wu, C., and Klenk, M. "Generative AI for product Design: Getting the right design and the design right". In: *Generative AI and HCI (GenAICHI) Workshop at CHI 2023*. 2023. DOI: 10.48550/arXiv.2306.01217.

[14] Filipowicz, A. L. S., Wu, C. C., Lee, M. L., Shamma, D. A., **Hakimi, S.**, Carter, S., Iliev, R., Harinen, T., Sumner, E., and Hogan, C. "Familiarity plays a unique role in increasing preferences for battery electric vehicle adoption". In: *Proceedings of the Annual Meeting of the Cognitive Science Society*. Vol. 44. 2022. URL: https://escholarship.org/uc/item/39w1q4rp.

[15] Harinen, T., Filipowicz, A., **Hakimi, S.**, Iliev, R., Klenk, M., and Sumner, E. "Machine learning reveals how personalized climate communication can both succeed and backfire". In: *CML4Impact Workshop at NeurIPS 2022*. 2022. DOI: 10.48550/arXiv.2109.05104.

[16] Zhang, Y., Chen, F., **Hakimi, S.**, Harinen, T., Filipowicz, A., Chen, Y., Iliev, R., Arechiga, N., Murakami, K., Lyons, K., Wu, C., and Klenk, M. "ConjointNet: Enhancing conjoint analysis for preference prediction with representation learning". In: *13th Multidisciplinary Workshop on Advances in Preference Handling (M-PREF 2022) at IJCAI 2022*. 2022. DOI: 10.48550/arXiv.2503.11710.

[17] Gopinath, D*, DeCastro, J*, Sumner, E, Morgan, A, **Hakimi, S**, Rosman, G, & Stent, S. "HMIway-env: A framework for simulating behaviors and preferences to support human-AI teaming in driving". In: *Human-centered Intelligent Services Safety and Trustworthiness (HCIS) Workshop at IEEE/CVF CVPRW 2022*. 2022. URL: https://openaccess.thecvf.com/content/CVPR2022W/HCIS/papers/Gopinath_HMIway-Env_A_Framework_for_Simulating_Behaviors_and_Preferences_To_Support_CVPRW_2022_paper.pdf.

[18] Sukumar, S., Ward, A. F., Elliott-Williams, C., **Hakimi, S.**, and Mozer, M. C. "Overcoming temptation: Incentive design for intertemporal choice". In: *Proceedings of the 3rd Multidisciplinary Conference on Reinforcement Learning and Decision Making*. 2017, pp. 11–14. DOI: 10.48550/arXiv.2203.05782.
Talk.

Conference Presentations

[1] Eum, B., Daviet, R., **Hakimi, S.**, and Webb, R. "Extending a neural random utility framework to explain why neuroforecasting works". Talk. Annual Meeting of the Society for Neuroeconomics, Pasadena, CA. Oct. 2026.
Best Talk Award.

[2] Nath, S. S.†, Jayawardana, V., Van, M., Klenk, M., and **Hakimi, S.** "Meaningful reward feedback interacts with strategy to steer creative decisions and maintain design diversity". Poster. Annual Meeting of the Society for Neuroeconomics, Pasadena, CA. Oct. 2026.

[3] Huang, R., Deng, W., Blevins, E., Srirangarajan, T., Wu, C., **Hakimi, S.**, Klenk, M., Wu, C. C., and Knutson, B. "Influence of price information on neuroforecasting aggregate demand for vehicles". Poster. Annual Meeting of the Society for Neuroeconomics, Pasadena, CA. Oct. 2026.

[4] **Hakimi, S.**, Madden-Rusnak, A.†, Van, M., Carter, S., Li, J., Filipowicz, A., and Klenk, M. "Psychological and physiological signals dissociate in ability to predict choice across the consumer journey". Poster. Annual Meeting of the Society for Neuroeconomics, Pasadena, CA. Oct. 2026.

[5] Eum, B., Daviet, R., **Hakimi, S.**, Knutson, B., and Webb, R. "AI-driven interpretable visual features for demand neuroforecasting". Poster. ISMS Marketing Science, Carcavelos, Portugal. June 2026.

[6] Blevins, E., Deng, W., Wu, C., Srirangarajan, T., **Hakimi, S.**, Klenk, M., Wu, C. C., and Knutson, B. "Neuroforecasting aggregate demand for vehicles". Poster. Annual Meeting of the Social and Affective Neuroscience Society, San Diego, CA. Apr. 2026.

[7] Eum, B., Knutson, B., Daviet, R., **Hakimi, S.**, and Webb, R. "AI-driven interpretable visual features for demand neuroforecasting". Talk. Annual Meeting of the Society for Neuroeconomics, Boston, MA. Oct. 2025.

[8] Deng, W. N., Srirangarajan, T., Klenk, M., Wu, C., Knutson, B., and **Hakimi, S.** "The brain leads while behavior lags: Neural activity forecasts market demand for vehicles while elicited preferences reflect past demand". Poster. Annual Meeting of the Society for Neuroeconomics, Boston, MA. Oct. 2025.

[9] **Hakimi, S.**, Christiano, D.†, Wu, C., Srirangarajan, T., Deng, W. N., Wu, C., Knutson, B., and Klenk, M. "Neuroforecasting versus backcasting changes in aggregate demand for vehicles". Poster. Annual Meeting of the Society for Neuroeconomics, Boston, MA. Oct. 2025.

[10] Christiano, D.†, Wu, C., Srirangarajan, T., Deng, N., **Hakimi, S.**, Klenk, M., Wu, C., and Knutson, B. "Neuroforecasting changes in aggregate demand for vehicles". Poster. 2025 Bay Area Affective Science Meeting, Stanford, CA. June 2025.

[11] **Hakimi, S.**, Ong, D., Gendron, M., and Kimani, E. "Interrogating the opportunities and risks of using generative AI for understanding and supporting affective processes". Symposium. Annual Meeting of the Society for Affective Science, Portland, OR. Apr. 2025.

[12] Christiano, D.†, Wu, C., Srirangarajan, T., **Hakimi, S.**, Klenk, M., Wu, C., and Knutson, B. "Neuroforecasting changes in aggregate demand for vehicles". Poster. Annual Meeting of the Society for Neuroeconomics, Cascais, Portugal. Oct. 2024.

[13] Christiano, D.†, Srirangarajan, T., **Hakimi, S.**, Klenk, M., Van, M., Wu, C., Murakami, K., and Knutson, B. "Brain responses to vehicles predict individual interest and forecast changes in aggregate demand". Poster. Annual Meeting of the Society for Neuroeconomics, Vancouver, BC. Oct. 2023.

[14] Hsiung, A.†, Pearson, J. M., Poh, J.-H., **Hakimi, S.**, Adcock, R. A., and Huettel, S. A. "Flexible integration of cost and evidence during information sampling". Poster. Annual Meeting of the Society for Neuroeconomics, Arlington, VA. Oct. 2022.

[15] Dickerson, K. C., Poh, J.-H., **Hakimi, S.**, Wright, R.†, Eom, K.†, Muzekari, B.†, Kollins, S., and Adcock, R. A. "Cognitive neurostimulation of dopaminergic midbrain via fMRI neurofeedback training increases willingness to exert effort in ADHD". Poster. Presented at the 60th Annual Meeting of the American College of Neuropsychopharmacology, Virtual. Program No. P122. Dec. 2021.

[16] **Hakimi, S.**, Muzekari, B. S.†, Eom, K. H.†, Shrestha, S.†, Wang, C., Ng, S., Rivera-Cancel, A. M., Thorp, J. N.†, Erwin, S. R., Wright, R. N.†, Zucker, N. L., and Adcock, R. A. "Dynamic changes in feedback valuation explain learning in a probabilistic two-armed bandit task". Presented at the Annual Meeting of the Society for Neuroeconomics, Virtual. Oct. 2020.

[17] Sinclair, A.†, **Hakimi, S.**, Stanley, M., and Samanez-Larkin, G. R. "Perceived versus actual virus transmission risk during the COVID-19 pandemic". Presented at the Annual Meeting of the Society for Neuroeconomics, Virtual. Oct. 2020.

[18] Choi, Y., Wright, R. N.†, **Hakimi, S.**, Hsiung, A.†, Sinclair, A.†, and Adcock, R. A. "Dissecting the relationship in information-seeking behaviors between clinically diagnosed anxiety and depression patients during the COVID-19 pandemic". Presented at the Duke Summer Neuroscience Program Poster Session, Virtual. Aug. 2020.

[19] Sinclair, A.†, **Hakimi, S.**, Adcock, R. A., and Barense, M. D. "Effective connectivity among cortico-hippocampal regions predicts memory for naturalistic episodes". Poster. Presented at the 2020 Context and Episodic Memory Symposium, Virtual. Aug. 2020.

[20] Muzekari, B. S.†, **Hakimi, S.**, Eom, K. H.†, Shrestha, S.†, Ng, S., Rivera-Cancel, A. M., Thorp, J. N.†, Erwin, S. R., Wright, R. N.†, Zucker, N. L., and Adcock, R. A. "Dynamic feedback valuation impacts learning in a probabilistic two-armed bandit task". Poster. Annual Meeting of the Cognitive Neuroscience Society, Virtual. May 2020.

[21] Eom, K. H.†, Rivera-Cancel, A. M., **Hakimi, S.**, Muzekari, B. S.†, Zucker, N. L., and Adcock, R. A. "Mapping the individual experience of perfectionism to clinical measures". Poster. Annual Meeting of the Society of Biological Psychiatry, Virtual. May 2020.

[22] Erwin, S. R., Ng, S., **Hakimi, S.**, Shrestha, S.†, Eom, K. H.†, Muzekari, B. S.†, Silberstein, K., Wright, R.†, Adcock, R. A., and Zucker, N. L. "Harnessing perfectionism: The role of emotion regulation and reward experience". Poster. 66th Annual Meeting of the American Academy of Child and Adolescent Psychiatry, Chicago, IL. Oct. 2019.

[23] Castrellon, J. J., Skene, J. H. P., Yin, L., **Hakimi, S.**, Parelman, J.†, Law, J. R., Skene, J. A. G., Ball, D., Malekpour, A., Pearson, J. M., and Carter, R. M. "Language and recall regions of the brain track evidence of guilt in mock criminal scenarios". Presented at the Annual Meeting of the Society for Neuroeconomics, Dublin, Ireland. Oct. 2019.

[24] **Hakimi, S.**, MacInnes, J. J., Dickerson, K. C., McDonald, K., and Adcock, R. A. "Encoded temporal representations predict VTA neurofeedback-mediated learning to self-regulate motivation". Presented at the 2019 Smokies Cognition and Neuroscience Symposium, Asheville, NC. Apr. 2019.

[25] **Hakimi, S.**, MacInnes, J. J., Dickerson, K. C., McDonald, K., and Adcock, R. A. "Encoded temporal representations predict VTA neurofeedback-mediated learning to self-regulate motivation". Presented at the 2019 DIBS Distinguished Lecture & Symposium, Durham, NC. Feb. 2019.

[26] Hsiung, A.†, **Hakimi, S.**, Adcock, R. A., Pearson, J. M., and Huettel, S. A. "Characterizing information sampling tendencies". Presented at the 2019 DIBS Distinguished Lecture & Symposium, Durham, NC. Feb. 2019.

[27] Thorp, J. N.†, **Hakimi, S.**, Dickerson, K. C., MacInnes, J. J., and Adcock, R. A. "Mesolimbic connectivity during neurofeedback training predicts learning to volitionally activate the ventral tegmental area". Poster. Presented at the Annual Meeting of the Society for Neuroscience, San Diego, CA. Program No. 334.19/III55. Nov. 2018.

[28] **Hakimi, S.**, MacInnes, J. J., Dickerson, K. C., and Adcock, R. A. "Encoded temporal representations predicting VTA neurofeedback-mediated learning to self-regulate motivation". Talk. Annual Meeting of the Society for Neuroeconomics, Philadelphia, PA. Oct. 2018.

[29] Hsiung, A.†, **Hakimi, S.**, Adcock, R. A., Pearson, J. M., and Huettel, S. A. "The effects of evidence accumulation on incidental memory". Presented at the Annual Meeting of the Society for Neuroeconomics, Philadelphia, PA. Oct. 2018.

[30] **Hakimi, S.**, MacInnes, J. J., Dickerson, K. C., and Adcock, R. A. "Temporal structure of learning to regulate ventral tegmental area using real-time neurofeedback". Poster. Conference on Cognitive Computational Neuroscience, Philadelphia, PA. Sept. 2018.

[31] Thorp, J. N.†, **Hakimi, S.**, Dickerson, K. C., MacInnes, J. J., and Adcock, R. A. "Mesolimbic connectivity during neurofeedback training predicts learning to volitionally activate the ventral tegmental area". Presented at the 4th Annual Meeting of the Triangle Chapter of the Society for Neuroscience, Cary, NC. May 2018.

[32] Dickerson, K. C., Biswas, S., Thorp, J. N.†, Shrestha, S.†, **Hakimi, S.**, and Adcock, R. A. "Neurofeedback of the dopamine system using EEG and fMRI". Presented at the Duke CTSA Symposium, Durham, NC. May 2018.

[33] Chandrashekharan, S. V.†, **Hakimi, S.**, Thorp, J. N.†, Shrestha, S.†, Dickerson, K. C., and Adcock, R. A. "The effects of labeling volitional motivation during real-time EEG neurofeedback". Presented at the Duke Undergraduate Neuroscience Graduation with Distinction Poster Session, Durham, NC. Apr. 2018.

[34] **Hakimi, S.**, MacInnes, J. J., Dickerson, K. C., and Adcock, R. A. "Modeling learning from real-time fMRI neurofeedback". Presented at the Third International Meeting on Real-Time Functional Imaging and Neurofeedback, Nara, Japan. Nov. 2017.

[35] **Hakimi, S.** "Neural correlates of cognitive control as a function of emergent automaticity". Presented at the Annual Meeting of the Society for Neuroeconomics, Toronto, Canada. Oct. 2017.

[36] **Hakimi, S.** "Neural correlates of cognitive control as a function of emergent automaticity". Poster. Multidisciplinary Conference on Reinforcement Learning and Decision Making, Ann Arbor, MI. July 2017.

[37] Biswas, S., **Hakimi, S.**, Kollins, S., Adcock, R. A., and Dickerson, K. C. "Evaluating real-time fMRI neurofeedback as an ADHD treatment". Presented at the Duke Psychology Vertical Integration Program Poster Session, Durham, NC. July 2017.

[38] **Hakimi, S.**, Clithero, J. A., Mullette-Gillman, O. A., Smith, D. V., McLaurin, R. E., Taren, A., Venkatraman, V., Huettel, S. A., and Carter, R. M. "Decomposing risk representation in parietal cortex". Presented at the 3rd Annual Meeting of the Triangle Chapter of the Society for Neuroscience, Cary, NC. Apr. 2017.

[39] Jung, H.†, Mosner, M. G., McLaurin, R. E., **Hakimi, S.**, Parelman, J. M.†, Kinard, J., Chakraborty, P., Dichter, G. S., and Carter, R. M. "MVPA testing for co-opted higher-order social cognition in autism". Poster. Annual Meeting of the Social and Affective Neuroscience Society, Los Angeles, CA. Mar. 2017.

[40] Carter, R. M., **Hakimi, S.**, Parelman, J. M.†, Law, J. R., Vidmar, N., Skene, J. A. G., Beskind, D. H., Pearson, J. M., and Skene, J. H. P. "The temporal parietal junction tracks evidence of guilt in mock criminal scenarios". Poster. Presented at the Annual Meeting of the Society for Neuroscience, San Diego, CA. Program No. 749.10/LLL35. Nov. 2016.

[41] **Hakimi, S.**, Clithero, J. A., Mullette-Gillman, O. A., Smith, D. V., McLaurin, R. E., Taren, A., Venkatraman, V., Huettel, S. A., and Carter, R. M. "Decomposing risk representation in parietal cortex". Poster. Annual Meeting of the Society for Neuroeconomics, Berlin, Germany. Aug. 2016.

[42] **Hakimi, S.**, Ward, A. F., Sukumar, S., Elliott-Williams, C., and Mozer, M. C. "Optimizing incentive design for intertemporal choice". Poster. Annual Meeting of the Society for Neuroeconomics, Berlin, Germany. Aug. 2016.

[43] Jung, H.†, Mosner, M. G., McLaurin, R. E., **Hakimi, S.**, Parelman, J. M.†, Kinard, J., Chakraborty, P., Dichter, G. S., and Carter, R. M. "Co-opted social cognitive neural substrates in autism during strategic gameplay". Poster. 71st Annual Scientific Convention and Meeting of the Society of Biological Psychiatry, Atlanta, GA. Abstract No. 202. May 2016.

[44] Jung, H.†, Mosner, M. G., McLaurin, R. E., **Hakimi, S.**, Parelman, J. M.†, Kinard, J., Chakraborty, P., Dichter, G. S., and Carter, R. M. "Testing for co-opted higher-order social cognition in autism". Talk. 86th Annual Convention of the Rocky Mountain Psychological Association, Denver, CO. Apr. 2016.

[45] Parelman, J. M.†, **Hakimi, S.**, Fairley, K., Foster, J. M., Jones, M., and Carter, R. M. "Risk preferences in a competitive social context". Poster. Presented at the Annual Meeting of the Society for Neuroeconomics, Miami, FL. Sept. 2015.

[46] Israelsen, B., Ward, A. F., Smith, I., Sukumar, S., **Hakimi, S.**, Lynch, J., and Mozer, M. "Overcoming temptation: Theory and practice". Presented at the Institute of Cognitive Science Colloquium, Boulder, CO. Apr. 2015.

[47] **Hakimi, S.**, Hare, T. A., and Rangel, A. "Variations in discount rates are associated with differences in the ability to imagine future rewards". Poster. Annual Meeting of the Society for Neuroeconomics, Evanston, IL. Oct. 2011.

[48] **Hakimi, S.**, Hare, T. A., and Rangel, A. "Variations in discount rates are associated with differences in the ability to imagine future rewards". Presented at the 2011 NSF IGERT Poster Competition. May 2011.

[49] **Hakimi, S.**, Hare, T. A., and Rangel, A. "Variations in discount rates are associated with differences in the ability to imagine future rewards". Presented at the Annual Meeting of the Cognitive Neuroscience Society, San Francisco, CA. Apr. 2011.

[50] **Hakimi, S.**, Hare, T. A., and Rangel, A. "Variations in discount rates are associated with differences in the ability to imagine future rewards". Presented at the NYAS Meeting on Critical Contributions of the Orbitofrontal Cortex to Behavior, New York, NY. Mar. 2011.

[51] Zink, C. F., Kempf, L., **Hakimi, S.**, Stein, J. L., and Meyer-Lindenberg, A. "Vasopressin effects on the neural correlates of social recognition". Poster. Presented at the Annual Meeting of the Society for Neuroscience, San Diego, CA. Program No. 403.3/LLL59. Nov. 2010.

[52] Tost, H., Braus, D. F., **Hakimi, S.**, Hohn, F., Ruf, M., and Meyer-Lindenberg, A. "Acute D2 receptor blockade induces rapid, reversible remodeling in human cortical-striatal circuits". Poster. Presented at the Annual Meeting of the Organization for Human Brain Mapping, Barcelona, Spain. Abstract No. 1401 WTh-AM. June 2010.

[53] Tost, H., Braus, D. F., **Hakimi, S.**, Hohn, F., and Meyer-Lindenberg, A. "A rapid, reversible neuroplasticity mechanism in humans reflects acute extrapyramidal side effects". Presented at the Annual Meeting of the American College of Neuropsychopharmacology, Hollywood, FL. Dec. 2009.

[54] Zink, C. F., Kempf, L., Stein, J. L., **Hakimi, S.**, and Meyer-Lindenberg, A. "Effects of vasopressin on mPFC-amygdala circuitry during fearful face processing in humans". Poster. Presented at the Annual Meeting of the Society for Neuroscience, Chicago, IL. Program No. 91.19/DD68. Oct. 2009.

[55] Zink, C. F., Kempf, L., **Hakimi, S.**, Rainey, C. A., Stein, J. L., and Meyer-Lindenberg, A. "Effects of vasopressin on mPFC-amygdala circuitry during fearful face processing in humans". Poster. Presented at the Annual Meeting of the Organization for Human Brain Mapping, San Francisco, CA. NeuroImage 47:S184. June 2009. DOI: 10.1016/S1053-8119(09)72042-7.

[56] Jansma, J. M., **Hakimi, S.**, Tong, Y., Zink, C. F., and Meyer-Lindenberg, A. "Trial by trial interaction effects in visual cortex and amygdala: Application of an fMRI method sensitive for non-linear effects". Poster. Presented at the Annual Meeting of the Society of Biological Psychiatry, Washington, DC. Abstract No. 155. May 2008.

[57] Jansma, J. M., **Hakimi, S.**, Fukunaga, M., Duyn, J. H., and Meyer-Lindenberg, A. "An fMRI study of trial-by-trial interaction effects in fusiform gyrus and amygdala". Talk. Presented at the Annual Meeting of the Society for Neuroscience, San Diego, CA. Program No. 669.4. Nov. 2007.

[58] **Hakimi, S.**, Kempf, L. P., Kolachana, B., Verchinski, B. A., Tong, Y., Mattay, V. S., Weinberger, D. R., and Meyer-Lindenberg, A. "fMRI investigation of the impact of genetic variation in Pten on the neural correlates of face emotion processing". Poster. Presented at the Annual Meeting of the Society for Neuroscience, San Diego, CA. Program No. 739.14/XX26. Nov. 2007.

[59] Zink, C. F., Tong, Y., Chen, Q., **Hakimi, S.**, Rainey, C., Stein, J., Laumann, T., Kempf, L., and Meyer-Lindenberg, A. "A comparison of the neural representation of social and non-social hierarchies in humans". Talk. Presented at the Annual Meeting of the Society for Neuroscience, San Diego, CA. Program No. 23.5. Nov. 2007.

[60] **Hakimi, S.**, Kempf, L. P., Kolachana, B., Stein, J. L., Verchinski, B. A., Mattay, V. S., Weinberger, D. R., and Meyer-Lindenberg, A. "Voxel-based morphometry investigation of the impact of genetic variation in Pten". Presented at the Annual Meeting of the Organization for Human Brain Mapping, Chicago, IL. June 2007.

[61] **Hakimi, S.**, Kempf, L. P., Kolachana, B., Stein, J. L., Verchinski, B. A., Mattay, V. S., Weinberger, D. R., and Meyer-Lindenberg, A. "Voxel-based morphometry investigation of the impact of genetic variation in Pten". Presented at the Postbaccalaureate IRTA Research Festival, Bethesda, MD. May 2007.

[62] Rainey, C. A., Zink, C. F., Draper, C. K., Chen, Q., Stein, J. L., Kempf, L., Tong, Y., Swaddle, J. P., **Hakimi, S.**, Laumann, T., and Meyer-Lindenberg, A. "Neural processing of facial cues of gonadal steroid status in humans". Poster. Presented at the Annual Meeting of the Society for Neuroscience, Atlanta, GA. Program No. 578.25/NN4. Oct. 2006.

[63] **Hakimi, S.**, Goldin, P. R., and Gross, J. J. "fMRI investigation of extraversion and positive self-referential processing in social phobia". Talk. Presented at the Annual Meeting of the Society for Neuroscience, Atlanta, GA. Program No. 311.10. Oct. 2006.

[64] **Hakimi, S.**, Goldin, P. R., Ramel, W. E., and Gross, J. J. "Neural mechanisms of emotion regulation in social anxiety disorder". Presented at the Annual Meeting of the Society of Biological Psychiatry, Toronto, Canada. May 2006.

[65] **Hakimi, S.**, Goldin, P. R., and Gross, J. J. "MRI investigation of positive self-referential processing in social anxiety disorder". Presented at the Annual Meeting of the Cognitive Neuroscience Society, San Francisco, CA. Apr. 2006.

[66] **Hakimi, S.**, Goldin, P. R., and Gross, J. J. "fMRI investigation of neuroticism and self-referential processing". Presented at the Annual Meeting of the Society for Personality and Social Psychology, Palm Springs, CA. Jan. 2006.

[67] Goldin, P. R., Anderson, B. A., **Hakimi, S.**, Hutcherson, C. A. C., and Gross, J. J. "Temporal dynamics of emotion responses: Affective chronometry of neural, autonomic and subjective emotion experience in response to sad and amusing films". Poster. Presented at the Annual Meeting of the Cognitive Neuroscience Society, New York, NY. Apr. 2005.

Reviews, Book Chapters & Commentaries

Commentaries

[1] Klenk, M., Hong, M., **Hakimi, S.**, and Wu, C. "Anticipatory thinking in design". In: *AI Magazine* 44.2 (2023), pp. 165–172. DOI: 10.1002/aaai.12101.

[2] Hsiung, A.[†], **Hakimi, S.**, and Adcock, R. A. "Unpacking the different flavors of curiosity". In: *IEEE CIS Newsletter on Cognitive and Developmental Systems* 15.1 (2018). Spring 2018, p. 10. URL: papers/Hsiung2018curiosity.pdf.

Book Chapters

[3] Tost, H., **Hakimi, S.**, and Meyer-Lindenberg, A. "Magnetic resonance biomarkers in schizophrenia research". In: *Handbook of neuropsychiatric biomarkers, endophenotypes, and genes: Promises, advances, and challenges*. Ed. by Ritsner, M. S. Dordrecht: Springer, 2009. Chap. 23, pp. 381–402. DOI: 10.1007/978-1-4020-9831-4_6.

[4] Tost, H., **Hakimi, S.**, and Meyer-Lindenberg, A. "Dopamine dysfunction in schizophrenia: From genetic susceptibility to cognitive impairment". In: *Dopamine Handbook*. Ed. by Iversen, L. and Iversen, S. Oxford University Press, 2009. Chap. 10.4, pp. 558–571. DOI: 10.1093/acprof:oso/9780195373035.003.0039.

Science Communication

[1] Bachman, M., Dickerson, K. C., **Hakimi, S.**, Li, R., and Yang, B. *How we formed a 'journal club' for equity in science*. 2020. DOI: 10.1038/d41586-020-02794-4.
Authors contributed equally and are listed alphabetically.

Patents

Granted

[1] **Hakimi, S.**, Lee, M. L., Wu, C. C., and Arechiga, N. "System and method for visualizing goals, relationships, and progress". US 12,775,888. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2026.

[2] Zhang, Y., Chen, F. R., Iliev, R., Harinen, T., Filipowicz, A. L. S., Chen, Y. Y., Arechiga, N., **Hakimi, S.**, Lyons, K. M., Wu, C. C., and Klenk, M. E. "Systems and methods providing a ConjointNet architecture for enhanced conjoint analysis for preference prediction with representation learning". US 12,737,593. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2026.

[3] Harinen, T., Iliev, R., **Hakimi, S.**, Filipowicz, A. L. S., and Sumner, E. S. "System and method for calculating generalized utilities and choice predictions". US 12,718,115. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2026.

[4] Hong, M. K., Toyoda, H., Chen, Y. Y., **Hakimi, S.**, and Klenk, M. E. "System and method for learning and communicating implicit stylistic preferences from historical user interaction data in text-to-image prompt engineering". US 12,613,892. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[5] Chen, Y. Y., **Hakimi, S.**, Van, M. P., and Hong, M. K. "System and method for inferring psychological complexity elicited by visual stimuli". US 12,541,651. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2026.

[6] Sumner, E. S., **Hakimi, S.**, Filipowicz, A. L. S., Carter, S., and Glazko, Y. S. "Systems and methods for encouraging application exploration". US 12,524,477. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[7] Zhang, Y., **Hakimi, S.**, and Klenk, M. E. "Systems and methods for detecting and regularizing drift in sequential decisions". US 12,524,132. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[8] Hong, M. K., **Hakimi, S.**, and Chen, Y. Y. "Automated moodboard augmentation via cross-modal generative association making". US 12,493,401. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[9] Shamma, D. A., Hogan, C., **Hakimi, S.**, and Weng, Y. "System and method for legitimate co-worker remote encounter interactions". US 12,425,255. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[10] **Hakimi, S.**, Wu, C. C., Lee, M. L., and Arechiga, N. "System and method for improving understanding and decision making regarding statistical data". US 12,400,077. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[11] Lee, M. L., **Hakimi, S.**, Arechiga, N., and Wu, C. C. "Systems and methods for investigating interactions between simulated humans". US 12,210,808. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2025.

[12] Sumner, E. S., Arechiga, N., Weng, Y., **Hakimi, S.**, and DeCastro, J. A. "Methods and systems for expert-novice guidance". US 12,169,697. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2024.

[13] Rosman, G., Brooks, D. J., Stent, S. A., Chen, T., Sumner, E. S., **Hakimi, S.**, and Gopinath, D. E. "Systems and methods for learning and managing robot user interfaces". US 12,084,080. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2024.

[14] Arechiga, N., Lee, M. L., Wu, C. C., and **Hakimi, S.** "System and method of a digital persona for empathy and understanding". US 12,062,121. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2024.

[15] Chen, Y. Y., **Hakimi, S.**, Lyons, K. M., Zhang, Y., Hong, M. K. S., Harinen, T., and Wu, C. C. "System and method to infer thoughts and a process through which a human generated labels for a coding scheme". US 12,032,618. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2024.

[16] **Hakimi, S.**, Wu, C. C., Lee, M. L., and Arechiga, N. "System and method for contextualizing and improving understanding of web search results". US 11,934,476. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2024.

[17] Carter, S., Filipowicz, A., **Hakimi, S.**, and Wu, C. "Ranked choice on an absolute scale". US 11,868,600. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2024.

[18] Arechiga, N., **Hakimi, S.**, Wu, C. C., and Lee, M. L. "System and method for an augmented reality goal assistant". US 11,579,684. Toyota Research Institute. U.S. Patent and Trademark Office, Washington, DC. 2023.

Filed

[1] Arechiga, N., Sumner, E. S., and **Hakimi, S.** "System and method for human-in-the-loop simulated annealing". US App. 18/913,880. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[2] Klenk, M. E., Nakanishi, R., **Hakimi, S.**, Yang, Y. Y., and Paredes, P. E. "System and method to create interactive personas for design". US App. 18/607,388. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[3] Jeon, Y., Hong, M. K., Chen, Y. Y., and **Hakimi, S.** "Systems and methods for manipulating images by comparing mapped styles and dimensions using machine learning models". US App. 18/539,748. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[4] Nakanishi, R., Chen, Y. Y., **Hakimi, S.**, Hong, M. K., and Klenk, M. E. "System and method to generate extended context derived from consumer responses". US App. 18/441,035. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[5] Nandy, A.[†], **Hakimi, S.**, Klenk, M. E., and Chen, Y. Y. "System and method for recommending design alternatives based on responses to semantic prompts". US App. 18/409,642. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[6] Stent, S., Best, A. P., **Hakimi, S.**, Rosman, G., Sumner, E. S., and DeCastro, J. A. "Adversarial simulation for developing and testing assistive driving technology". US App. 18/156,591. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[7] **Hakimi, S.**, Wu, C. C., and Zhang, Y. "System and method for learning relationships between physiological signals and psychological state to predict individual behavior". US App. 18/099,187. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[8] DeCastro, J. A., Rosman, G., Stent, S. A. I., Sumner, E. S., **Hakimi, S.**, Gopinath, D. E., and Morgan, A. "System and method for training at least one policy using a framework for encoding human behaviors and preferences in a driving environment". US App. 18/098,776. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[9] **Hakimi, S.**, Carter, S., DeCastro, J. A., Sumner, E. S., Weng, Y., and Arechiga, N. "System and method for machine-assisted collaboration in product design". US App. 17/550,496. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

[10] Weng, Y., Sumner, E. S., **Hakimi, S.**, Arechiga, N., and DeCastro, J. A. "Systems and methods for prototype generation". US App. 17/507,187. Toyota Research Institute. U.S. Patent Application, Patent and Trademark Office, Washington, DC.

Invited Talks

West Coast Neuroeconomics Mini-Symposium, University of California, Los Angeles — Los Angeles, CA
"Psychologically-guided generative AI interventions for flexible, creative decision making." — 2025

West Coast Neuroeconomics Mini-Symposium, University of California, San Diego — San Diego, CA
"Abstract semantic prompts shape sequential decision making in design." — 2025

2025 Annual Convention of the Western Psychological Association — Las Vegas, NV
"Neuroforecasting market demand for electrified vehicles: Leveraging the neuroscience of affect and motivation to predict future consumer behaviors." — 2025

"Rethinking How 'Design Spaces' are Constructed" Workshop, Design Computing & Cognition (DCC) '24 — Montréal, Canada
"Resolving mental simulation of possible futures for understanding and supporting design." — 2024

Berkeley Institute of Design Seminar, University of California, Berkeley — Berkeley, CA
"Rethinking user understanding and applications for innovation in design." — 2024

Department of Psychology, Temple University — Philadelphia, PA
"Learning to self-regulate motivation using real-time fMRI neurofeedback." — 2018

Department of Economics, University of Zürich — Zürich, Switzerland
"Learning to regulate motivation using real-time neurofeedback." — 2018

Center for Cognitive Neuroscience Colloquium, Duke University — Durham, NC
"The academic, there and back again." — 2019

Central Institute of Mental Health (ZI) — Mannheim, Germany
"Modeling the neural basis of individual differences in intertemporal choice." — 2016

Department of Computer Science, University of Colorado Boulder — Boulder, CO
"Neural correlates of intertemporal choice." — 2015

Teaching

Duke University — Durham, NC
Guest Lecturer — Fall 2018

- DECSCI 101 / PSY 141: Fundamentals of Decision Science, "Motivation and Decision Making"

California Institute of Technology — Pasadena, CA
Guest Lecturer & Curriculum Designer — Fall 2011

- CNS/Bi/Ph 107: Writing About Scientific Research

California Institute of Technology — Pasadena, CA
Teaching Assistant — Fall 2009

- Bi/CNS 150: Introduction to Neurobiology

Wellcome Trust Conference Center — Hinxton, Cambridge, UK
Teaching Assistant — Summer 2009

- Wellcome Trust School on the Biology of Social Cognition

Professional Service & Outreach

Society for Neuroeconomics — Victoria, BC, Canada
Board Member *(elected)* — 10/2026 – present

Women & Allies Employee Resource Group (WAERG), Toyota Research Institute — Los Altos, CA
Co-Chair *(elected)* — 03/2022 – 03/2024

Project Invent — San Francisco, CA
Idea Review Mentor & Pitch Coach — 10/2021 – 04/2022

Center for Cognitive Neuroscience, Duke University — Durham, NC
Postdoctoral Representative *(nominated & elected)* — 08/2019 – 01/2021

Inclusion and Power Dynamics in Academia Series, Duke University — Durham, NC
Co-Founder & Leadership Committee Member — 11/2017 – 01/2021

- Collaborator, Duke Faculty Advancement Seed Grants (2018–2020)

SPEAK: Scientists Promoting Equity and Knowledge, Duke University — Durham, NC
Co-Founder & Leadership Committee Member — 06/2016 – 01/2021

***PNAS* Front Matter, National Academy of Sciences** — Washington, DC
Journal Club Panel Member — 06/2018 – 08/2020

Organization for Human Brain Mapping — Minneapolis, MN
Communications Committee Member & Student/Postdoc SIG Liaison — 01/2017 – 12/2018

Upper school course: *Decision Making*, Carolina Friends School — Durham, NC
Guest Instructor & Curriculum Designer — 03/2017 – 05/2017

Reel Science at Caltech, California Institute of Technology — Pasadena, CA
Presenter — 03/2012 – 04/2013

Behavioral Health & Ortho/Neuro Trauma Units, Suburban Hospital — Bethesda, MD
Clinical Volunteer — 10/2006 – 07/2007

Reviewing

Journals

Acta Psychologica; Cognitive, Affective, and Behavioral Neuroscience; Computational Psychiatry; Experimental Brain Research; Journal of Child Psychiatry and Psychology; Journal of Experimental Psychology: Learning, Memory, and Cognition; Memory and Cognition; Nature Human Behaviour; New Ideas in Psychology; NeuroImage; PLoS One; Psychological Science; Social, Cognitive, and Affective Neuroscience

Conferences

ACM CHI Conference on Human Factors in Computing Systems (2026); ACM Creativity and Cognition Conference (C&C, 2026); Cognitive Science Society (2025–2026); Generative AI and HCI Workshop at CHI (GenAICHI, 2025); Organization for Human Brain Mapping (OHBM; 2010–2011, 2016–2019); Society for Neuroeconomics (SNE, 2025–2026); Triangle Society for Neuroscience (SfN, 2018); Workshop on Behavioral ML at NeurIPS (2024)

Professional Memberships

Association for the Advancement of Artificial Intelligence; Cognitive Science Society; Society for Affective Science; Society for Judgment and Decision Making; Society for Neuroeconomics